清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

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Assessment of Low-Carbon Technologies in the Iron and Steel Industry

Tsinghua University & Saudi Aramco

aramco

清华大学气候变化与可持续发展研究院
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Project Timeline

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	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July	Aug
Research Proposal & preparations																
Data requirement & database investigation																
Case study: Saudi Arabia & USA																
Case study: Japan & Germany & Australia																
Case study: Russia & India									1							
Overview of global steel trade & technology assessment																
Accounting models & technology report																
High spatial resolution data collation & Model development																
Data analysis & Scenario design																
Final report																

The iron & steel industry is a key area of global emission reduction

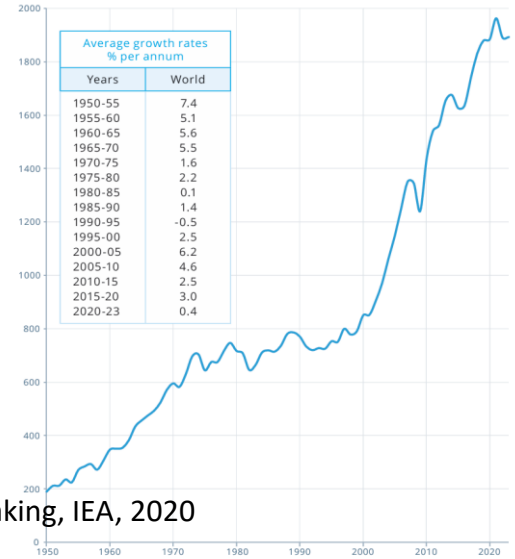
- The steel industry accounts for approximately 7-9% of global carbon dioxide emissions[1]
- The steel industry accounts for about 8% of global energy demand[2]
- The energy consumption structure of the steel industry is dominated by coal[3]
- Even under the most ambitious decarbonization scenarios, world steel demand is projected to grow by more than 30% by 2050[3]
- Nearly 20 countries outside China have steel production capacity of 10 million tons or close to it. Their steel production emissions reduction also needs attention.

STEEL

Key performance data 2023[291,292,293]

7%	2.8 Gt CO_2e	1.8%
Contribution to global CO_2e emissions	Scope 1 and 2 emissions	Emissions increase (2019-2023)
1.5 tCO_2e/t	83%	1.3 times
Emissions intensity (per tonne of steel, 2023)	Fossil fuels in the fuel mix (2022)	Demand in NZE scenario by 2050, compared to 2023

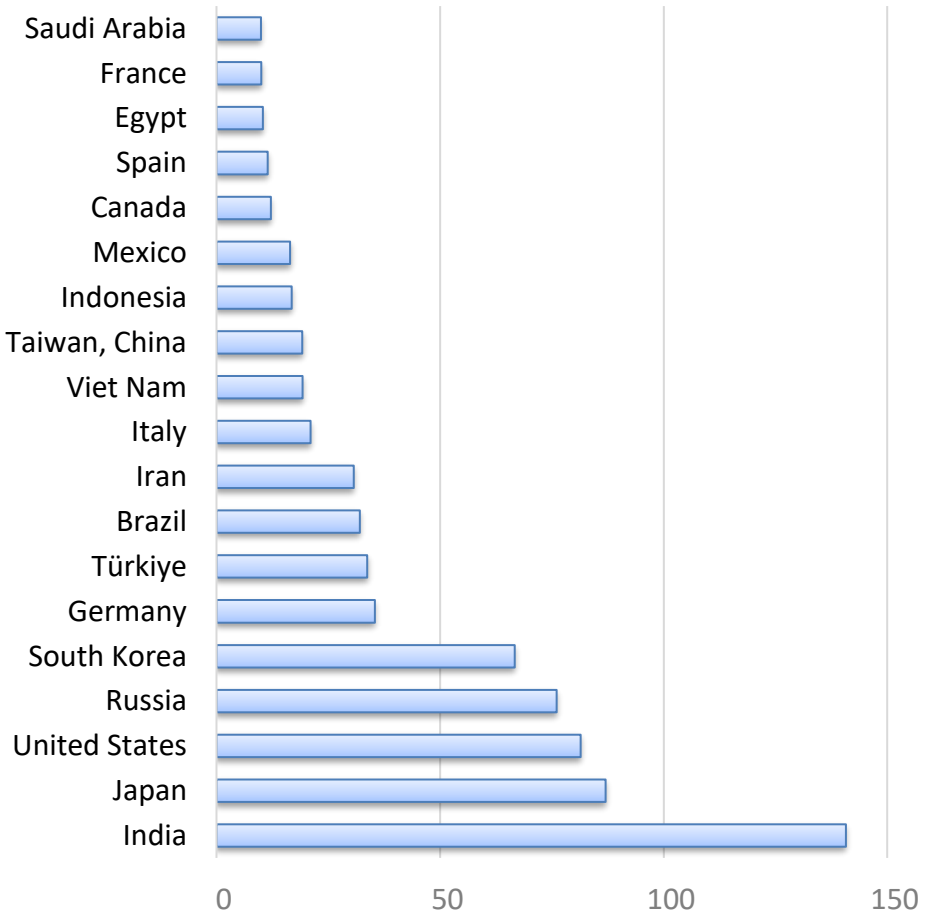
million tonnes, crude steel production



Average growth rates % per annum

Years	World
1950-55	7.4
1955-60	5.1
1960-65	5.6
1965-70	5.5
1970-75	1.6
1975-80	2.2
1980-85	0.1
1985-90	1.4
1990-95	-0.5
1995-00	2.5
2000-05	6.2
2005-10	4.6
2010-15	2.5
2015-20	3.0
2020-23	0.4

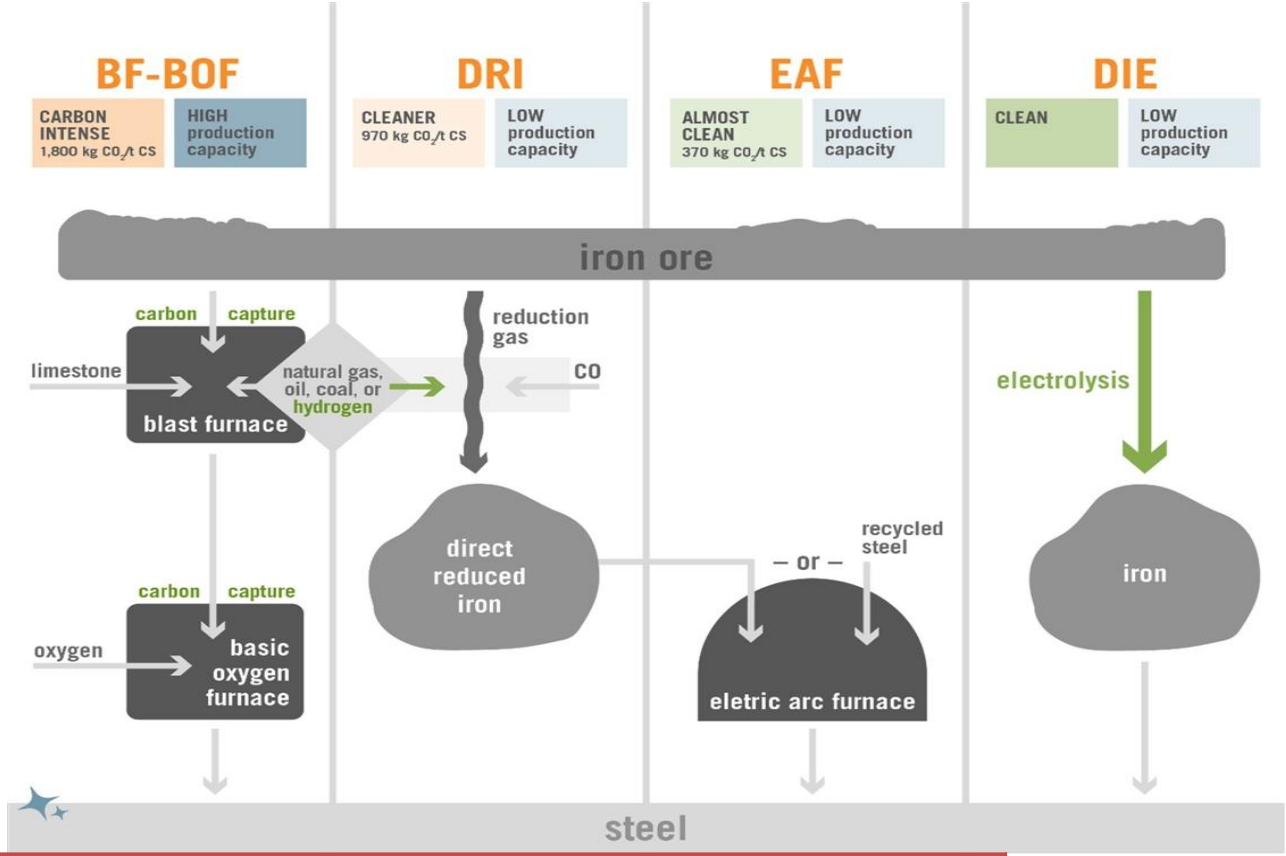
Crude steel production by country in 2023 (Mt)



[1] World Steel in Figures 2024, Worldsteel, 2024
[2] Iron and Steel Technology Roadmap - Towards more sustainable steelmaking, IEA, 2020
[3] Net-Zero Industry Tracker: 2024 Edition, WEF, 2024

Main technological routes of the worldwide steel industry

- **BF-BOF**: Traditional long-process steelmaking technology route
- **Scrap-EAF**: Steelmaking technology route that mainly utilizes scrap steel - **some DRI is added to improve purity**
- **DRI-EAF**: A short-process steelmaking technology route using direct reduced iron; **coal**, **natural gas**, or **green hydrogen** can be used as reducing agents - some scrap steel is added to save costs
- **BF-OHF**: The outdated technology route using **open-hearth furnaces** (its technological share is extremely low)
- **DRI-IF**: Using induction Furnace generated by electromagnetic induction; generally only used in extremely small steel mills (micro-mills)
- **DIE**(Direct Iron Ore Electrolysis): Dissolving iron ore in a specialized molten electrolyte for electrolytic reduction; still in the laboratory research and development stage.



Technology process	reductant	Direct & indirect CO2 emission per crude steel (tCO_2/tSteel)	Energy Consumption (GJ/t)	Share of global steel production (%)
BF-BOF	Coke	1.7-2.5	24.2	73.2
Scrap-EAF	/	0.35-0.66	10.24	21.5
DRI-EAF	NG	1.5-1.7	17.1	3.8
	Coal	2.5-3.2	23.13	1.0

What are DRI products?

	CDRI	HBI	HDRI
Fe Total (%)	90 - 94	90 - 94	90 - 94
Fe Metallic (%)	83 - 90	83 - 90	83 - 90
Metallization (%)	92 - 97	92 - 96	92 - 96
Carbon (%)	1.0 - 4.0	0.5 - 3.0	1.0 - 4.0
P* (%)	0.005 - 0.09	0.005 - 0.09	0.005 - 0.09
S* (%)	0.001 - 0.03	0.001 - 0.03	0.001 - 0.03
Gangue* (%)	2.8 - 6.0	2.8 - 6.0	2.8 - 6.0
Bulk Density (kg/m³)	1,600 - 1,900	2,400 - 2,800	1,600 - 1,900
Bulk Density (lbs/ft³)	100 - 120	150 - 175	100 - 120
Product Temperature (℃)	50	80	600 - 700
Typical Size (mm)	4 - 20	30 x 50 x 110	4 - 20

DRI Product	Manufacture	Transport	Storage
CDRI	The product formed by the **direct cooling** of HDRI	● Water (Inland): Must be protected from rain & moisture ● Water (Ocean): Holds must be kept inert per IMO guidelines ● Rail & Road: Must be protected from rain & moisture	● Silos & Bins: Must be kept cool & dry ● Piles: Must be covered with proper drainage
HDRI	The **direct product** formed after iron ore is reduced	● Localized transport only: Sealed conveyors or hot transport vessels; product is of most value when used shortly after production at or nearby steel mill	● Not applicable
HBI	The product formed by **briquetting HDRI** (under pressure) at a high temperature of 650° C	● Water (Inland): No special precautions ● Water (Ocean): No special precautions ● Rail & Road: No special precautions	● Silos & Bins: No special precautions ● Piles: Uncovered with good drainage

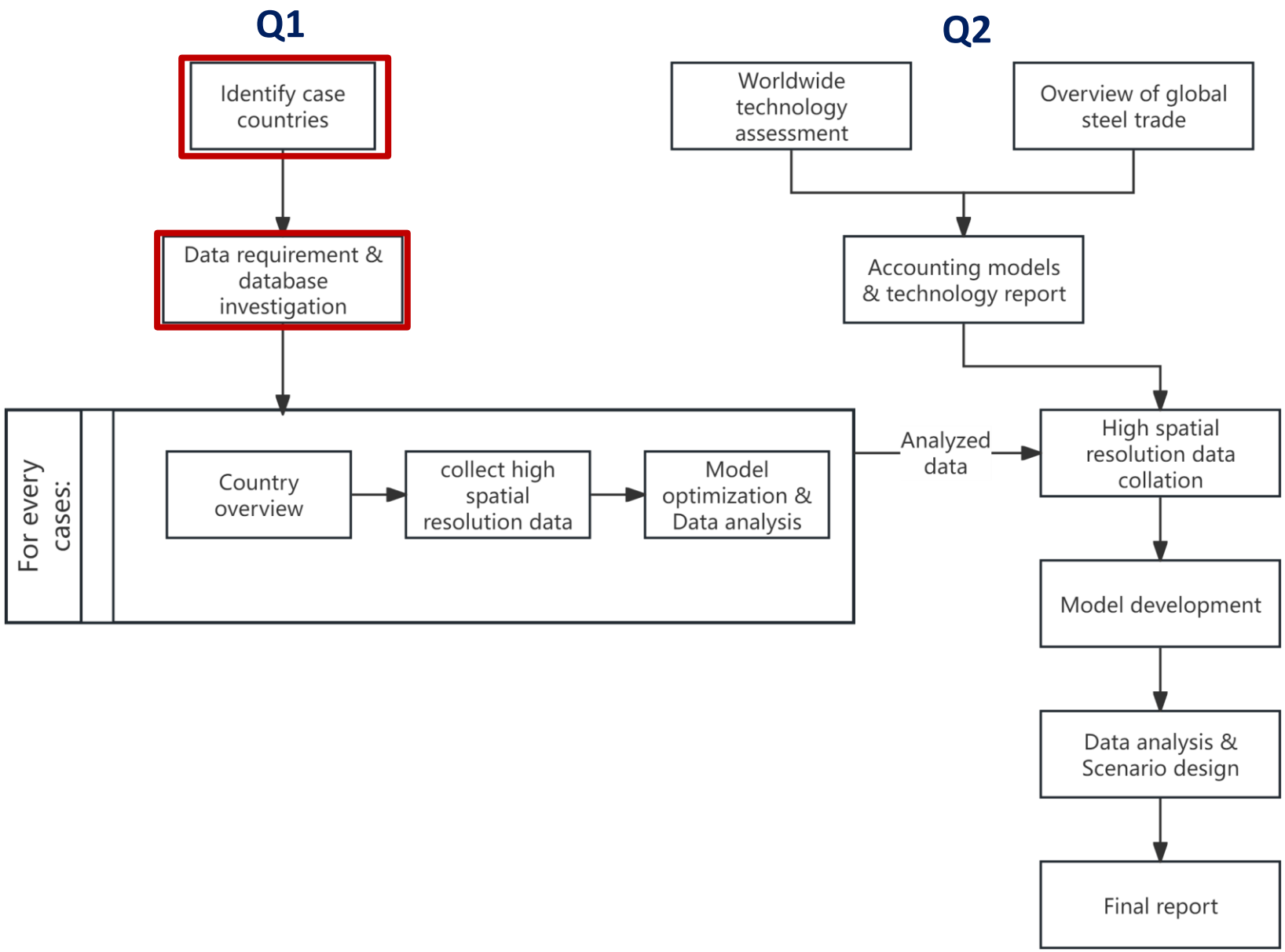
Source: https://www.midrex.com/wp-content/uploads/MIdrexDRI_ProductsBrochure_4-12-18.pdf

Research Questions

- **Q1： What are the economic and emission reduction benefits of DRI in the world's major steel-producing countries?**
  - Different case countries
    - iron ore resource endowments
    - energy costs, labor costs, hydrogen production costs
    - market environments, steel demand, ...
  - We only use a techno-economic model to calculate costs, but other variables (such as industrial base, infrastructure conditions, labor quality, etc.) should also exist that would affect the actual DRI cost.
  - **A study about Technology Suitability, which need more case studies & Model optimization (if necessary)**

- **Q2： Under the current international trade policy environment and energy transformation trends, what impact might this new steel technology application have on global steel trade?**
  - Emerging H2-DRI technology and trade models based on DRI products could potentially impact traditional iron ore trade.
  - International trade policies such as US tariffs and the European CBAM (Carbon Border Adjustment Mechanism) will impact steel production and the division of labor in supply chains across various countries.

Research roadmap



Identify case countries: some key indicators

Country	International division of Labor			Technology mix		DRI Production (Mt)	Energy Cost	
	Crude steel production(Mt)	Iron ore production(Mt)	Net Iron ore imports (Mt, import-export)	BOF rate(%)	EAF/IF rate(%)		Wind Power CAPEX(USD/kW)	Solar PV CAPEX(USD/kW)
Saudi Arabia	9.9(20th)	0.0	43.68	0	100	6.8	~1611***	794
Austrailia	5.5(28th)	944.1(1st)	-887.0	73.2	26.8	/	~1610***	989
Japan	87.0(3rd)	0.0	104.2	73.8	26.2	/	~2019***	/
USA	81.4(4th)	46.0	-6.5	31.7	68.3	5.2	~1484***	1109
India	140.8(2nd)	247.8(4th)	-14.0	43.6	56.4	49.6	1208	711
Germany	35.4(7th)	0.8	36.0	72.3	27.7	0.2	~1583***	731
Russia	76.0(5th)	90.9	-22.9*	65.1**	32.0**	7.8	~1611***	1248

Year: **2023**; Data sources: **Worldsteel, IRENA, World Bank, Rosstat**

* Worldsteel provided data for Russia & other CIS + Ukraine: ore production volume 174.8 Mt, exports 44.1 Mt, imports 0; Rosstat provided Russia's ore production volume as 90.9 Mt; figures were derived by proportional scaling.

** Russia still has open-hearth furnace steelmaking, which accounts for the remaining 2.9% of its technological structure.

*** Wind power installation cost data is provided based on the country's region.

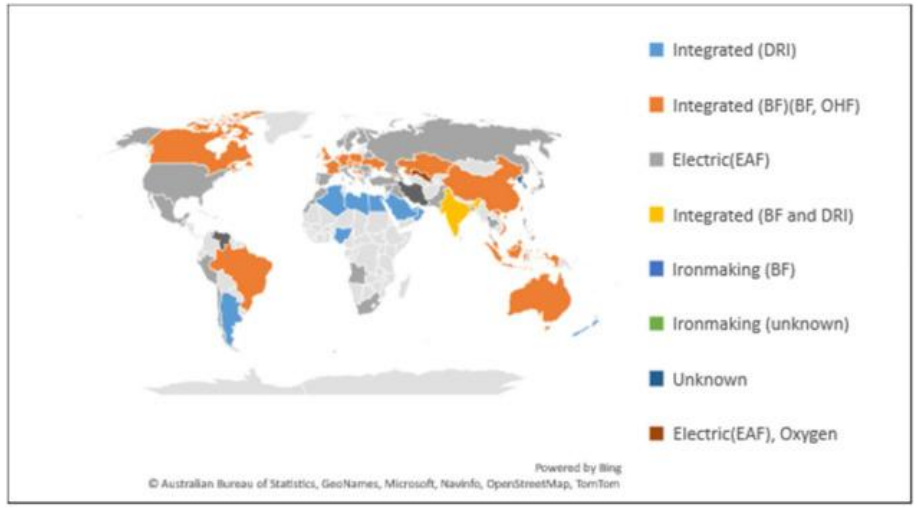
Identify case countries

Division by Technological Structure

Technological Structure	Description	Representative Countries
Electric (EAF)	Scrap-EAF Dominant	USA
Integrated (DRI)	Gas-DRI-EAF Dominant, generally due to low natural gas prices	Saudi Arabia, Egypt
Integrated (BF)	BF-BOF Dominant	Germany, Brazil, Australia, Japan
Integrated (BF and DRI)	proportions of BF-BOF and Coal-DRI-EAF/IF routes are comparable	India

Division by Global Steel Trade Status

Global Steel Trade Status	Description	Representative Countries
Mining	Mines iron ore, which is mostly used for export	Australia, Brazil
Smelting	Mainly relies on imports to obtain iron ore, which is then smelted into steel products	China, Japan, Saudi Arabia
Mining and Smelting	Mainly relies on domestic mining to obtain iron ore, which is then smelted into steel products	Russia, USA, India



	Mining	Smelting	Mining and Smelting
Electric (EAF)	/	Mexico	**USA**
Integrated (DRI)	/	**Saudi Arabia,** Egypt	/
Integrated (BF)	**Australia,** Brazil, South Africa	China, Korea, **Germany**, **Japan**	**Russia**
Integrated (BF and DRI)	/	/	**India**

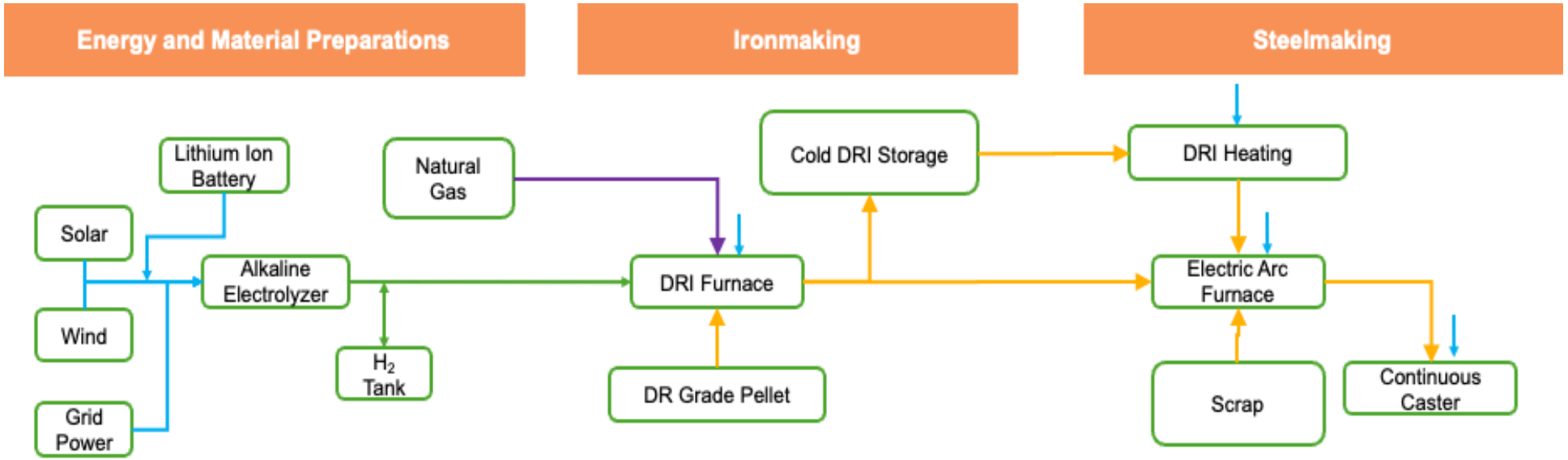
BARARZADEH LEDARI M et el. Greening steel industry by hydrogen: Lessons learned for the developing world[J]. International Journal of Hydrogen Energy, 2023, 48(94)

The present H_2-DRI-EAF model

■ Flexibility Analysis of H_2-DRI-EAF system 氢能钢铁灵活生产系统建模分析

An **hourly resolution system optimization model** is developed, integrating electricity, hydrogen, iron, and steel production, storage, and consumption. The model evaluates the flexible operational modes of different production units and assesses the overall economic and environmental benefits of the system. This study unveils a paradigm shift in the temporal decoupling and flexible operation of production processes under renewable-centric H_2-DRI-EAF.

氢能钢铁灵活生产系统建模：建立了考虑电、氢、铁、钢的生产、储存、消费的小时级系统优化模型，评估了不同生产设备的灵活运行模式和系统整体的经济与环境效益，揭示了基于可再生绿氢DRI电炉炼钢技术的生产流程之间的时域解耦和灵活运行的范式转变的特征。



Data requirements (model H2-DRI-EAF)

- **DRI Location**: also where solar PV & wind power site
- **Maximum Renewable Output(kWh/kWp)**: hourly, Solar & Wind, determined by weather conditions
- **Steel demand**: annual
- **Industrial Equipment Cost:** solar, wind, battery power, battery energy, electrolyzer, H_2 tank, DRI furnace, EAF, annuitized cost including CAPEX and fixed OPEX
- **Raw material cost**: Iron ore, Scrap
- **Traditional energy cost**: Grid Power, Natural Gas
- **Labor cost**: Ironmaking labor, Steelmaking labor
- **Stoichiometry parameters & Carbon emission intensity**: same as China's scenario

Database: DRI location

- steelonthenet.com (SOTN): Proprietary
  - Plant-level capacity data, technology & location specific, build-up and retirement year (for each unit)
  - Scope: Crude steel and its upstream (coke, sinter, pig iron/DRI, crude steel)

6	30	20	6	20	20	30	8	8	8	8	8	8	8	8	8	8
18 JANUARY 2024																
Table																283
IRON AND STEEL CAPACITY BY PLANT AND BY COMPANY - ALL PRODUCTS, PLANTS AND COUNTRIES																
(000 tonnes per year capacity at 31 December 2023, ownership at date of update)																
Owner	ner Na	ner Cou	Site	Plant Region	Plant Country	Plant Name	coke	sinter	pig iron			DRI	crude steel			
									BF iron	ther pig iro	total pig iron		EAF steel	BOF steel	OHF steel	total crude
			1423	North America	United States	Charter Specialty Steel					0.0		640.6			640.6
			1450	North America	United States	Citizens Gas & Coke	668.8				0.0					0.0
			85	North America	United States	Cleveland Cliffs - Indiana Harbor		1263.7	3930.4		3930.4			4422.8		4422.8
			1570	North America	United States	Cleveland-Cliffs	579.2				0.0					0.0
			28	North America	United States	Cleveland-Cliffs Burns Harbor	1790.2	2112.6	5388.7		5388.7			5898.8		5898.8
			83	North America	United States	Cleveland-Cliffs Cleveland			3071.7		3071.7			3644.6		3644.6
			1334	North America	United States	Cleveland-Cliffs Coatesville					0.0		1102.7			1102.7
			15769	North America	United States	Cleveland-Cliffs DRI					0.0	1615.6				0.0
			1519	North America	United States	Cleveland-Cliffs Indiana Harbor		1276.9	7802.5		7802.5		526.5	5265.2		5791.7
			13	North America	United States	Cleveland-Cliffs Riverdale					0.0			1158.4		1158.4
			1569	North America	United States	Cleveland-Cliffs South Chiacgo	647.6				0.0					0.0
			1663	North America	United States	Cleveland-Cliffs Steelton					0.0		1338.4			1338.4
			15636	North America	United States	CMC Steel					0.0		467.3			467.3
			1734	North America	United States	CMC Steel South Carolina					0.0		883.6			883.6
			14695	North America	United States	CMC Steel South West					0.0		862.2			862.2
			15202	North America	United States	Columbus Steel Castings					0.0		235.2			235.2
			1314	North America	United States	Commercial Metals					0.0		1115.5			1115.5
			1315	North America	United States	Commercial Metals					0.0		497.0			497.0
			1456	North America	United States	Commercial Metals					0.0		1158.4			1158.4
			1458	North America	United States	Crucible Materials					0.0		74.3			74.3
			1459	North America	United States	CSC Industries					0.0		727.8			727.8
			1465	North America	United States	Edgewater Steel					0.0		43.8			43.8
			1466	North America	United States	Electralloy					0.0		63.2			63.2
			1583	North America	United States	Ellwood National Steel					0.0		131.2			131.2
			1467	North America	United States	Ellwood Quality Steels					0.0		315.9			315.9

Database: Maximum Renewable Output

- Renewables.ninja : Open Source
  - Maximum Solar PV Output calculated from MERRA-2 weather dataset (2019), with system loss rate, tilt and azimuth angle given
  - Maximum Wind power Output calculated from MERRA-2 weather dataset (2019), with hub height and Turbine model (manufacturer, blade diameter, and the rated capacity, exp. "Vestas V112-3.0MW") given
  - Scope: Worldwide, hourly data in 2019

# Renewables.nir	49.653 - Version: 1.1 (using	
# Units: time in U	local_time in Asi	electricity ir
# {"units": {"time"	"local_time": "As	"electricity "
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2019/1/1 0:00	2019/1/1 3:00	0
2019/1/1 1:00	2019/1/1 4:00	0
2019/1/1 2:00	2019/1/1 5:00	0
2019/1/1 3:00	2019/1/1 6:00	0.006
2019/1/1 4:00	2019/1/1 7:00	0.196
2019/1/1 5:00	2019/1/1 8:00	0.419
2019/1/1 6:00	2019/1/1 9:00	0.579
2019/1/1 7:00	2019/1/1 10:00	0.673
2019/1/1 8:00	2019/1/1 11:00	0.706
2019/1/1 9:00	2019/1/1 12:00	0.67
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2019/1/1 13:00	2019/1/1 16:00	0.022
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2019/1/1 17:00	2019/1/1 20:00	0
2019/1/1 18:00	2019/1/1 21:00	0
2019/1/1 19:00	2019/1/1 22:00	0
2019/1/1 20:00	2019/1/1 23:00	0
2019/1/1 21:00	2019/1/2 0:00	0
2019/1/1 22:00	2019/1/2 1:00	0
2019/1/1 23:00	2019/1/2 2:00	0
2019/1/2 0:00	2019/1/2 3:00	0
2019/1/2 1:00	2019/1/2 4:00	0
2019/1/2 2:00	2019/1/2 5:00	0

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2019/1/1 1:00	2019/1/1 4:00	0.394
2019/1/1 2:00	2019/1/1 5:00	0.345
2019/1/1 3:00	2019/1/1 6:00	0.319
2019/1/1 4:00	2019/1/1 7:00	0.261
2019/1/1 5:00	2019/1/1 8:00	0.174
2019/1/1 6:00	2019/1/1 9:00	0.108
2019/1/1 7:00	2019/1/1 10:00	0.07
2019/1/1 8:00	2019/1/1 11:00	0.03
2019/1/1 9:00	2019/1/1 12:00	0.016
2019/1/1 10:00	2019/1/1 13:00	0.029
2019/1/1 11:00	2019/1/1 14:00	0.068
2019/1/1 12:00	2019/1/1 15:00	0.091
2019/1/1 13:00	2019/1/1 16:00	0.134
2019/1/1 14:00	2019/1/1 17:00	0.165
2019/1/1 15:00	2019/1/1 18:00	0.13
2019/1/1 16:00	2019/1/1 19:00	0.068
2019/1/1 17:00	2019/1/1 20:00	0.021
2019/1/1 18:00	2019/1/1 21:00	0.006
2019/1/1 19:00	2019/1/1 22:00	0.007
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2019/1/1 23:00	2019/1/2 2:00	0.112
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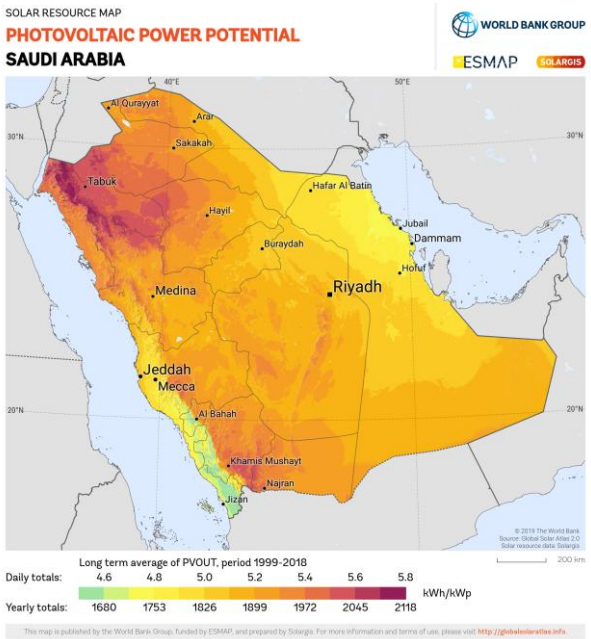
2019/1/2 1:00	2019/1/2 4:00	0.223
2019/1/2 2:00	2019/1/2 5:00	0.278
2019/1/2 3:00	2019/1/2 6:00	0.279
2019/1/2 4:00	2019/1/2 7:00	0.212

Hourly PVOut & Wind capacity factor data - Jubail & Dammam, Saudi-Arabia
from renewables.ninja



PVOut - Saudi Arabia(kWh/kWp)
from: Global solar atlas

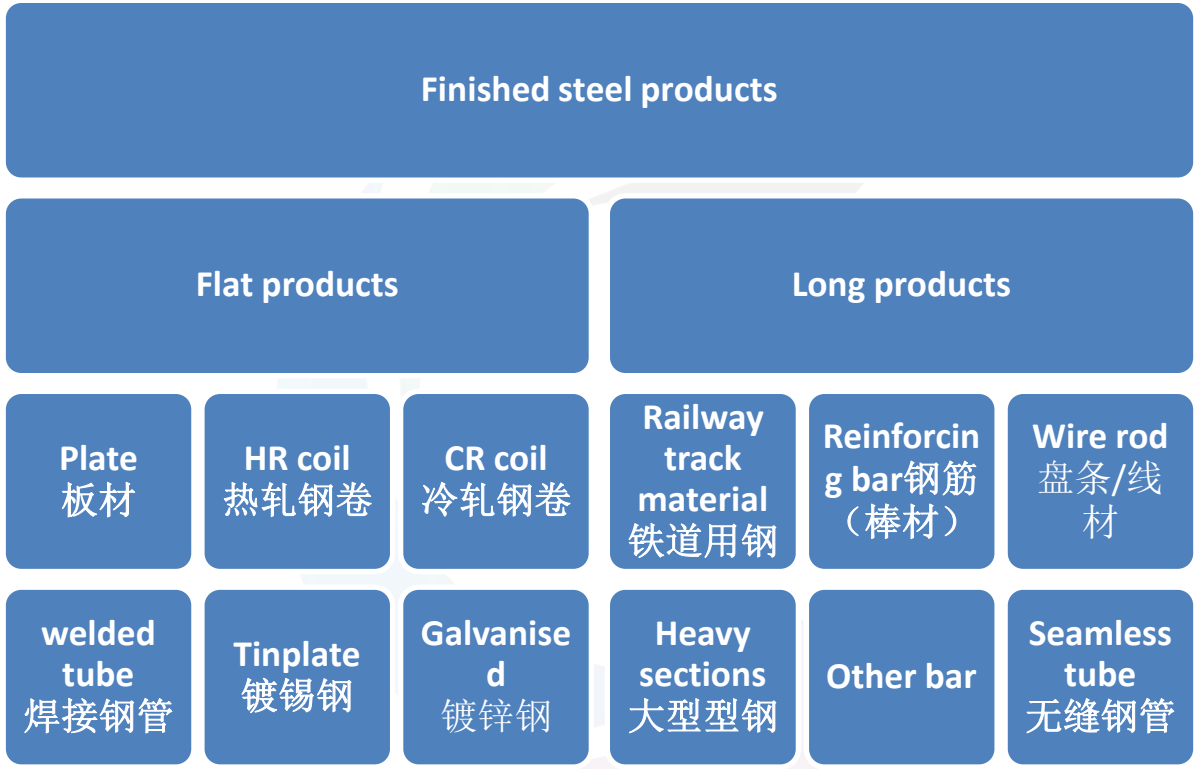
Capacity factor - Saudi Arabia(kWh/kWp)
from: Global wind atlas

Database: Steel demand

- steelonthenet.com (SOTN): Proprietary
  - Consumption data from 1970~2050, including predictions, national-level
  - Import & export data from 1970~2050

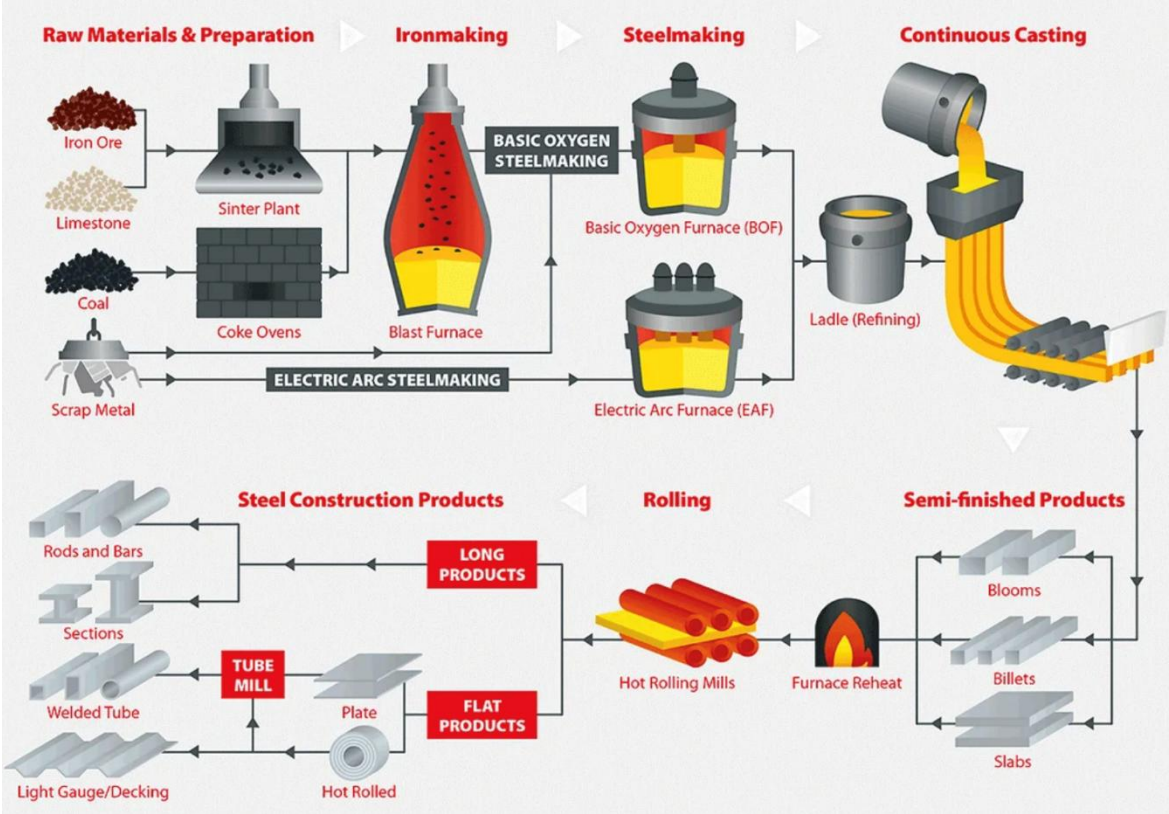
FINISHED STEEL CONSUMPTION - TOTAL
(000 tonnes)

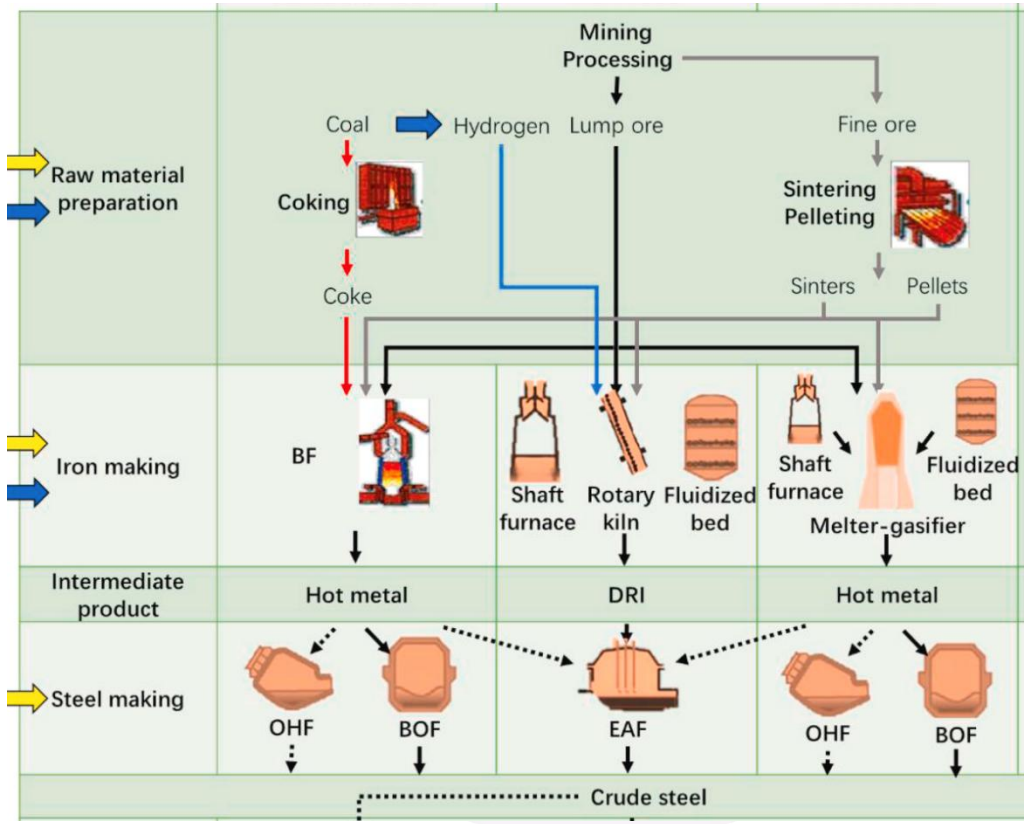
Region	Country	Product	Item	16	2017	2018	2019	2020	2021	2022	2023
North America	Canada	Finished stee	Consumption	.7	12725.4	15377.9	15951.1	13166.0	13604.2	14888.5	14092.4
North America	United States	Finished stee	Consumption	.5	97380.2	99399.1	97284.2	80156.0	99275.5	87410.5	91841.9
Latin America	Argentina	Finished stee	Consumption	.0	5002.6	4639.0	4144.2	3528.6	5019.4	3868.5	2875.7
Latin America	Bolivia	Finished stee	Consumption	.2	735.9	721.0	736.0	509.9	804.0	570.6	559.2
Latin America	Brazil	Finished stee	Consumption	.7	21970.3	24554.6	23081.2	22672.6	28839.1	25464.0	25765.4
Latin America	Chile	Finished stee	Consumption	.9	2668.6	3076.4	2935.8	2455.9	3306.7	1981.2	1961.9
Latin America	Colombia	Finished stee	Consumption	.0	4121.8	4172.6	4131.5	3053.5	4124.6	3979.7	3171.9
Latin America	Costa Rica	Finished stee	Consumption	.3	736.3	788.5	697.2	561.3	743.2	665.5	700.3
Latin America	Cuba	Finished stee	Consumption	.0	299.4	269.0	268.8	150.5	167.9	194.5	216.0
Latin America	Dominican Re	Finished stee	Consumption	.2	775.3	853.1	959.0	751.2	983.0	602.0	678.3
Latin America	Ecuador	Finished stee	Consumption	.1	1650.1	1657.5	1665.5	1264.2	1721.1	1606.6	1667.2
Latin America	El Salvador	Finished stee	Consumption	.6	405.0	420.1	417.8	370.4	549.4	338.5	378.6
Latin America	Guatemala	Finished stee	Consumption	.7	844.9	831.9	932.6	851.7	1177.4	880.7	961.5
Latin America	Honduras	Finished stee	Consumption	.9	433.4	473.9	436.4	373.3	544.0	178.6	212.0
Latin America	Jamaica	Finished stee	Consumption	.5	176.2	151.4	198.5	174.2	237.4	85.3	83.2
Latin America	Mexico	Finished stee	Consumption	.4	25796.2	26275.5	24221.4	21799.0	26085.0	23650.8	23957.2
Latin America	Nicaragua	Finished stee	Consumption	.7	280.7	212.6	249.3	280.2	364.1	183.2	201.1
Latin America	Panama	Finished stee	Consumption	.9	514.4	460.5	378.7	169.7	338.7	342.7	338.8
Latin America	Paraguay	Finished stee	Consumption	.5	358.3	446.2	348.0	302.7	388.3	304.5	384.3
Latin America	Peru	Finished stee	Consumption	.5	2730.5	2989.7	3178.3	2536.1	3730.7	2119.3	2459.3
Latin America	Trinidad	Finished stee	Consumption	.1	212.2	207.6	183.4	146.6	174.4	225.0	243.2
Latin America	Uruguay	Finished stee	Consumption	.1	202.8	211.3	191.6	234.2	279.2	292.9	252.4
Latin America	Venezuela	Finished stee	Consumption	.0	555.2	153.3	153.5	120.6	170.9	256.8	243.8
Latin America	Other Latin A	Finished stee	Consumption	.3	393.9	414.5	473.0	497.5	481.2	486.1	455.7
Europe	Albania	Finished stee	Consumption	.6	311.9	356.5	373.4	178.4	257.0	190.6	219.6
Europe	Austria	Finished stee	Consumption	.9	4455.8	4510.6	4313.9	3957.3	4871.0	4544.6	3463.4



Database: price

- steelonthenet.com (SOTN): Proprietary
  - Price data: 1985~2050
  - Scope: Thermal coal, Metallurgical coal, Metallurgical coke; Iron ore sinter fines, **Iron ore pellet fines**, lump (BF/**DR**), **Scrap**, Pig iron, DRI, steel products. Free on board & cost and freight





https://www.vepica.com/blog/steel-production-from-iron-ore-to-functional-industrial-products

Database: Labor cost

- Literature review: **Global green hydrogen-based steel opportunities surrounding high quality renewable energy and iron ore deposits**, table S18-19
  - Calculating Steel Labor Costs Using Average wage
  - **Steelworker wage = Average wage * 1.3**
  - hourly labor cost = **Steelworker wage / (1 + Tax Rate)**

- Database: World Bank (Open source)
  - Gross National Income (GNI) per capita = Average wage

Table S18 – Regionally variable steelworker wages

	Average wage (before tax)	Steelworker wage (before tax)		Personal Income Tax Rate (highest)	Steelworker hourly rate, inc. tax
Source	[47]	30% increase above average [2]		[48]	
Unit	$/year	$/year	$/hr	% of income	$/hr
AU	53620	69706	36.31	45%	52.64
BR	7820	10166	5.29	28%	6.75
CA	43540	56602	29.48	33%	39.21
CL	12990	16887	8.80	40%	12.31
CN	10520	13676	7.12	45%	10.33
GN	960	1248	0.65	40%	0.91
IN	1890	2457	1.28	43%	1.83
IR	3290	4277	2.23	35%	3.01
KZ	8710	11323	5.90	10%	6.49
MX	8750	11375	5.92	35%	8.00
NZ	41480	53924	28.09	39%	39.04
PE	6000	7800	4.06	30%	5.28
RU	10740	13962	7.27	13%	8.22
ZA	6090	7917	4.12	45%	5.98
SE	54740	71162	37.06	52%	56.34
TR	9070	11791	6.14	40%	8.60
UA	3570	4641	2.42	18%	2.85
US	64650	84045	43.77	37%	59.97

Table S19 – Labour intensities

Process	Labour intensity	Source
Green H2 production	2 hours/kW installed electrolysers	Own calculations using data from Leguijt, et al. [49], considering labour demand for RE generation and electrolysis parts of value chain.
DR ironmaking	0.22 hours/t DRI	Own calculations using data from Wood, et al. [2].
EAF steelmaking	0.49 hours/t steel	

Thank You!